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Unit – 2 : Session – 6 : SLO – 1 | Memory Locations and Addresses (1's & 2's Complement)

■ Section A – Match the Following (Answers)

1. 1's Complement → B. Flip all bits (0→1, 1→0)
2. 2's Complement → A. Add 1 to the 1's complement
3. Signed Number Representation → C. Represents negative numbers in binary
4. Most Significant Bit (MSB) → D. Indicates sign in signed numbers
5. Overflow → E. Error when result exceeds representable range

■ Section B – Fill in the Blanks

- 1's complement of a binary number is obtained by FLIPPING ALL BITS.
- 2's complement of a number is obtained by taking the 1's complement and then ADDING 1.
- In 2's complement representation, a negative number always begins with 1 in the MSB.
- The range of an 8-bit 2's complement number is from -128 to +127.
- When the result exceeds the possible range of values, it causes an OVERFLOW.

■ Section C – Activity Task

- Task 1 – 1's and 2's complements:
 - a) 0110 1001 → 1's: 1001 0110, 2's: 1001 0111.
 - b) 1011 0000 → 1's: 0100 1111, 2's: 0101 0000.
 - c) 0001 1110 → 1's: 1110 0001, 2's: 1110 0010.
- Task 2 – Convert to decimal (2's complement method):
 - 1111 0100 → negative; 1's = 0000 1011, 2's = 0000 1100 = 12 → value = -12.
 - 1000 1111 → negative; 1's = 0111 0000, 2's = 0111 0001 = 113 → value = -113.

■ Section D – True or False

- 1's complement is obtained by adding 1 to the binary number. — FALSE
- 2's complement is the most widely used method for representing signed numbers. — TRUE
- In 2's complement, there are two representations for zero. — FALSE (only one)
- MSB determines the sign of a signed number. — TRUE
- Taking 2's complement twice returns the original number. — TRUE